

Intelligent Financial Fraud Detection — Technical Proposal

NTU Datathon 2026 · Track 2 (Finance) · Team: The Larpers

1. Problem understanding

Binary classification of 20,000 labelled transactions with only **1.76% fraud** (353 cases). Accuracy is useless (always "legit" = 98%). We optimise the ranking quality (**PR-AUC**) and pick a decision threshold that balances catching fraud (**Recall**) against disrupting legitimate customers (precision), summarised by **F1**.

2. Data exploration — key insights

- **Velocity & device:** ≥ 4 transactions in the last hour $\rightarrow$ 10–30% fraud; new device 7.5% vs 1.2%; both together **~37%** (strong interaction).
- **Amount:** fraud rate rises from $\sim 1\%$ (<200) to 5–7% (500–3,000).
- **Account age:** accounts younger than 6 months are $\sim 3\times$ riskier.
- **Context:** luxury, cash transfer and electronics are the riskiest categories; non-SG countries are $\sim 2.5\times$ riskier; night hours (23h–4h) are slightly riskier.
- **Data quality:** ~ 0.5 –1% missing values in 9 columns (train and test); `id` dropped.
- **Distribution shift:** the test set has higher amounts, older accounts and more new devices $\rightarrow$ we favour a model that generalises over one that fits train perfectly.

3. Preprocessing & feature engineering

Missing categoricals $\rightarrow$ explicit "missing" level; missing numerics $\rightarrow$ handled natively by gradient boosting (median + indicator for linear models). 12 engineered features capture *behavioural deviation*: log amount, amount vs. 24h average ticket, amount / 24h spend, burst ratio (1h/24h), new-device $\times$ burst, new-account flag, foreign flag, night flag and cyclic hour ($\sin/\cos$). Categoricals are one-hot encoded with unseen levels ignored, which keeps the model robust to edge cases.

4. Models compared (5-fold stratified CV, out-of-fold)

Model	PR-AUC	ROC-AUC	Best F1
Random guess	0.018	0.500	—
Logistic Regression (baseline, class-weighted)	0.185	0.775	0.281
Random Forest (balanced, min_leaf=20)	0.178	0.763	0.267
HistGradientBoosting, shallow (selected)	0.200	0.767	0.282

Deeper / default-depth boosters (LightGBM, XGBoost, deep HGB) scored lower in CV: they overfit the noisy minority class.

5. Final model & why

HistGradientBoostingClassifier with depth 3, learning rate 0.02, 300 iterations, min 80 samples per leaf and $L2 = 5$. We chose it because:

- It captures the non-linear interactions found in EDA.
- It handles NaNs natively.
- It is insensitive to skewed scales and outliers.
- Strong regularisation helps it generalise under distribution shift.

The decision threshold is tuned with internal 5-fold CV (`TunedThresholdClassifierCV`) to maximise $F_{1.5}$, weighting recall slightly above precision because a missed fraud costs more than a review. The tuned threshold is ≈ 0.067 , versus a default of 0.5 that would flag almost nothing. It is stored inside the model, so `predict()` returns labels and `predict_proba()` gives scores for PR-AUC.

Honest estimate (nested CV, threshold tuning included): PR-AUC 0.20 (11 $\times$ random) · Recall 0.27–0.28 · F1 0.25 · Precision 0.23. In other words, $\sim 2\%$ of transactions are flagged and they contain $\sim 28\%$ of all fraud. Top drivers by permutation importance are velocity (1h/24h), new device, amount and account age.

6. Assumptions

- Transactions are i.i.d.: there is no customer ID or timestamp, so we use a stratified rather than temporal split.
- Labels are correct.
- The 1h/24h behavioural aggregates are computed before the transaction, so there is no leakage.
- Fraud prevalence in deployment is similar to training.

7. Limitations & improvements

The signal is noisy, so precision at useful recall is modest; the model is best used to rank a human-review queue rather than to auto-block. Improvements: per-customer history and time-based validation (if IDs and timestamps were available), a cost-sensitive threshold using real fraud and false-positive costs, probability calibration, drift monitoring with periodic threshold recalibration, and richer signals (device fingerprint, IP, merchant history).